

Exact equation

Consider the first-order equation

$$M(x,y) + N(x,y) \cdot y' = 0.$$

If there is a C^2 -function F such that

$$\frac{\partial F}{\partial x} = M \quad \text{and} \quad \frac{\partial F}{\partial y} = N$$

then, for a solution $y = \varphi(x)$,

$$\begin{aligned} M(x,y) + N(x,y) \cdot y' &= \frac{\partial F}{\partial x}(x, \varphi(x)) + \frac{\partial F}{\partial y}(x, \varphi(x)) \cdot \varphi'(x) \\ &= \frac{d}{dx} F(x, \varphi(x)) = 0. \end{aligned}$$

So, we obtain the implicitly relation

$$F(x, \varphi(x)) = C.$$

Moreover, by the Fubini theorem, if $M_y = N_x$, we can be guaranteed the existence of such a F . It is necessary-sufficient condition.

Further, although $M_y \neq N_x$, if there is a function $\mu(x,y)$ such that

$$\frac{\partial}{\partial y} [\mu(x,y) \cdot M(x,y)] = \frac{\partial}{\partial x} [\mu(x,y) \cdot N(x,y)]$$

$$\Leftrightarrow \mu_y \cdot M + \mu \cdot M_y = \mu_x \cdot N + \mu N_x$$

$$\Leftrightarrow \mu_x \cdot N - \mu_y \cdot M = (M_y - N_x) \cdot \mu. \quad \dots (*)$$

So, (*) is necessary-sufficient condition for such μ exists.

In general, the condition (*) is too hard to solve, (it is PDE)

But, if $(M_y - N_x)/N$ is a function dependent to only the variable x , then the condition (*) implies

$$\frac{1}{\mu} d\mu = \frac{M_y - N_x}{N} dx \Rightarrow \mu = \exp \left(\int \frac{M_y - N_x}{N} dx \right)$$

Problem.

Consider

$$(x \cdot e^{\frac{y}{x}} + 2 \cdot x^2 y) dy - y \cdot e^{\frac{y}{x}} dx = 0.$$